FloodNet NYC - Data Description

General

Dataset Name: FloodNet: Street Flooding Events Measured by FloodNet Sensors

Agency Name: New York City Department of Environmental Protection

Update Frequency: Biweekly

Dataset Description: Data presented here consist of flood event data measured by FloodNet sensors. Each row represents one flood event as measured by one sensor, and includes event summary information including start and end times of the flood, flood depth, and duration. The time series of flood depth data between the start and end times of the event is also included in array format.

The data were collected by the FloodNet project, which is a collaboration between researchers at academic institutions (New York University and the City University of New York) and NYC government agencies (NYC Department of Environmental Protection, Mayor's Office of Climate & Environmental Justice, Office of Technology & Innovation), working with NYC residents. Data on flood water depth were collected by FloodNet sensors in one minute intervals, then analyzed to develop the flood event summary statistics presented here. A flood event is defined as a series of water depth measurements greater than 10 mm; all measured flood events presented here have undergone QC by a member of the FloodNet team.

Dataset Keywords: flooding, flood depth, flood sensors

Dataset Category: Environment

If you have questions about the dataset, please contact info@floodnet.nyc

What is FloodNet NYC?

FloodNet NYC is a university/city agency consortium tasked with installing 500 flood sensors across the city's five boroughs by mid 2027, primarily funded by the New York City Department of Environmental Protection (NYC DEP). Our mission is to develop tools for real-time urban flood monitoring, implement these tools to measure flooding in New York City, and make flood data and monitoring tools available in a manner that is accessible and useful to stakeholders including residents, community-based organizations, government agencies, and researchers. The following

sections describe one of our event based data products. For information on how our sensors measure flood depth, see this section below).

Data Summary

This document will describe flood data as “flood event data” and “time series depth data”. There is an additional sensor metadata file (“Floodnet: Sensor Deployment Metadata.csv”) that provides individual sensor metadata such as its location and unique ID.

The data presented on Open Data are organized into “flood event data”, where each row of the Flood Events Table contains information on one flood event measured by one sensor. The “time series depth data” are the individual flood depth measurements collected by the sensor at specified time intervals. A flood event is defined as a series of depth measurements greater than 10 mm; all events have been reviewed by both an automated data analysis pipeline and a human review process to check that the measured event is a flood and not caused by changes in depth due to other technical or environmental factors. The automated data analysis process is applied every time the detected depth beneath a sensor exceeds 10 mm, and uses a Recurrent Neural Network (RNN) machine learning model, trained on a dataset of human-labeled flood and non-flood depth measurements collected by FloodNet sensors, to assign a “flood”/”non-flood” label to each “flood event data” row. A human then verifies each RNN-classified “flood” event, based on a visual review of the event graph alongside supplemental precipitation and/or tide level data (from non-FloodNet sources). Only human verified “flood” events are included in the dataset.

The “Data-Dictionary: Flood Events” attachment to this dataset provides a description of each flood event metric and how they are measured. Fig 1. shows a diagram of the “time series depth data” with select “flood event data” fields labelled.

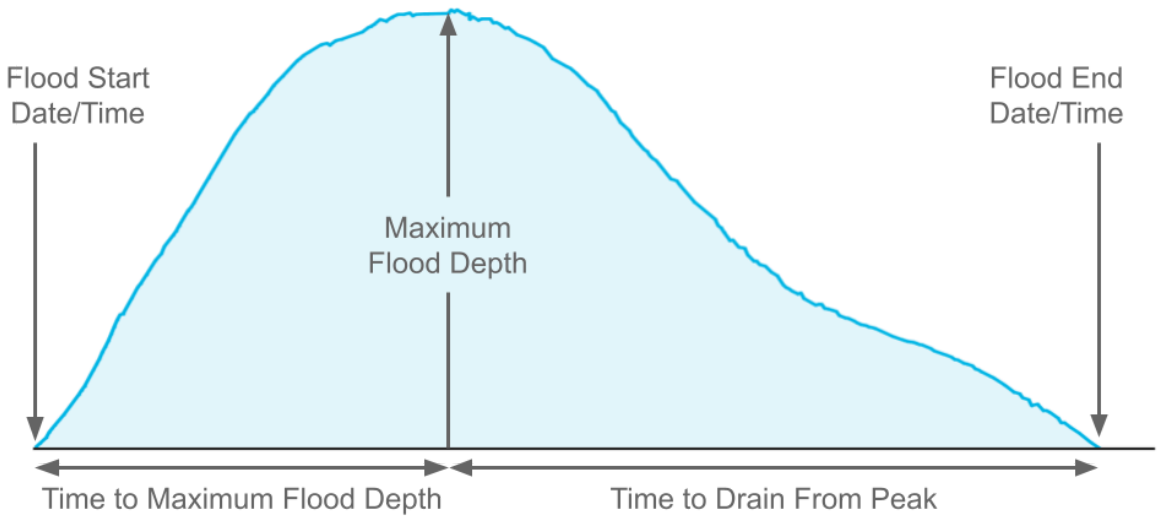


Fig. 1 - Time series depth data with key event metrics labelled

The sensor metadata includes fields that can be used to associate FloodNet sensor metadata with Flood Events. The "Sensor ID" field in each dataset should be used to make the association. The “Data-Dictionary: Floodnet - Sensor Deployment Metadata.xls” file (found in the NYC Open Data Attachments list for this dataset) contains descriptions of each field.

The sensor network grew in size from its first sensor deployment in October 2020 with sensors being retired for various reasons. The “Date Installed” and “Date Removed” fields provide a sensor’s deployment period. A sensor with an empty “Date Removed” field means it is currently deployed. The “Data-Dictionary: Floodnet - Sensor Deployment Metadata.xls” file is updated at the same frequency as this dataset (biweekly) as new sensors are deployed and/or others retired.

A field of note in the sensor metadata is “Sensor Ground Height Above Local Low Point (inches)” which is the difference in elevation, in inches, between the surface beneath the sensor (i.e., the surface being monitored for flooding) and the lowest point in the immediate local area, where flood water is expected to accumulate first. Given limitations in street infrastructure available for installing the sensors, sensors are typically mounted over a section of street or sidewalk that is not at the lowest elevation, where measured flood depth would be greatest, in that immediate local area. The "sensor height above local low point" enables the estimation of flood depth at the deepest point in the immediate local area.

Citation

If using the data or including it in any publication or public presentation, the project should be listed as “FloodNet (New York University and The City University of New York)”, and cited as the following:

> Mydlarz, C., Sai Venkat Challagonda, P., Steers, B., Rucker, J., Brain, T., Branco, B., Burnett, H.E., Kaur, A., Fischman, R., Graziano, K., Krueger, K., Henaff, E., Ignace, V., Jozwiak, E., Palchuri, J., Pierone, P., Rothman, P., Toledo-Crow, R., Silverman, A.I. (2024) FloodNet: Low-cost ultrasonic sensors for real-time measurement of hyperlocal, street-level floods in New York City. *Water Resources Research*, 60, e2023WR036806. https://doi.org/10.1029/2023WR036806

How do the sensors measure flood depth?

FloodNet sensors use ultrasonic pulses to **measure the distance** from their mounting position to the horizontal surface below the sensor at a regular interval, currently **once every 60 seconds**. Some sensors operating before November 2021 were programmed to measure distance once every 5 minutes, therefore some data before November 2021 may be presented with a time interval > 1 minute. In addition, occasional transmission issues and noise filtering processes to remove invalid measurements (see this section below) may result in some time series depth data with intervals > 1 minute.

The sensors work using ultrasound: they emit ultrasonic sound waves, outside the frequency range of human hearing, and then record the time it takes for the wave to bounce back off the closest reflective surface and return to the sensor. The sensor then uses the travel time of the ultrasonic pulse to calculate the distance to that reflective surface (z), whether it is the ground or water surface, using Equation 1, where v is the speed of sound and Δt is the time that it takes for the sound to reflect back and return to the sensor (divided by two because the ultrasonic pulse is traveling twice the distance we're measuring). The sensor has an internal temperature sensing component and corrects v for the air temperature, given the temperature-dependence of the speed of sound.

$$z = v\frac{\Delta t}{2} \qquad \text{Eq. 1}$$

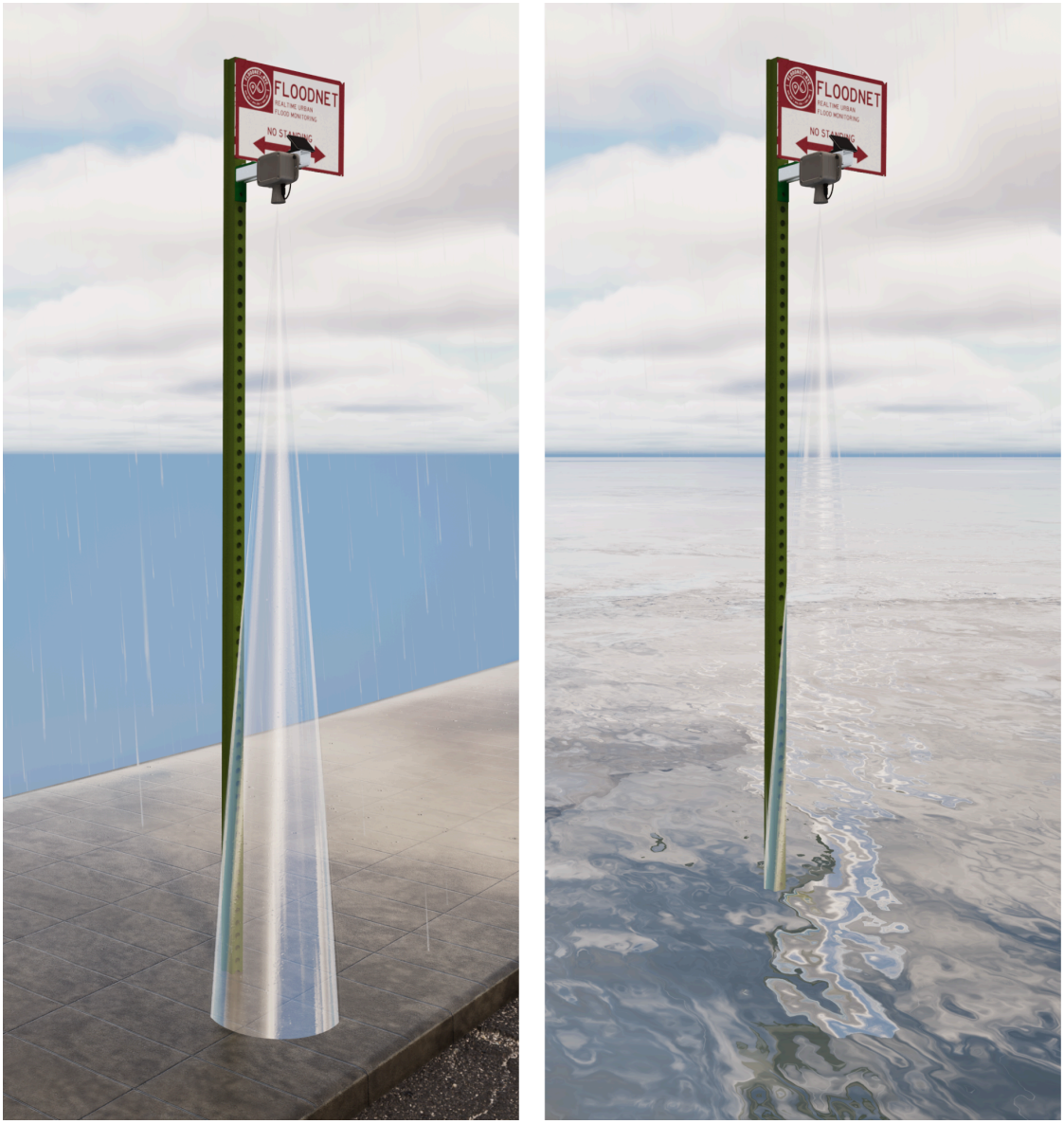


Fig. 2 - Digital mockup of FloodNet sensor operation, showing ultrasonic pulse with its spread during transmission in left image. Right image shows flood conditions beneath the sensor.

To calculate flood depth, the measured distance between the sensor and the ground below must be known under non-flooded conditions. This value is calculated through a dynamic calibration procedure that occurs at 5 AM daily, in which the distance measurements (z) collected over the previous three nights (between 10 PM and 5 AM) are analyzed to determine the median value ($z_{night\sim median}$). If the standard deviation of the night-time distance measurements exceeds 10 mm (signifying either a flood or erroneous high variance) or if the new calibration deviates excessively from the expected sensor height, the previous day's $z_{night\sim median}$ calculation is used. To calculate flood depth (D_t) at time point t, the sensor's distance measurement at that time (z_t) is subtracted from $z_{night\sim median}$ (Equation 2).

$$D_t = z_{night\sim median} - z_t \qquad \text{Eq. 2}$$

This dynamic calibration approach allows data collection to adapt to potential changes in sensor height or seasonal variation in baseline $z_{night\sim median}$ readings.

More information on the flood sensor technology and data processing can be found at:

Mydlarz, C., Sai Venkat Challagonda, P., Steers, B., Rucker, J., Brain, T., Branco, B., Burnett, H.E., Kaur, A., Fischman, R., Graziano, K., Krueger, K., Henaff, E., Ignace, V., Jozwiak, E., Palchuri, J., Pierone, P., Rothman, P., Toledo-Crow, R., Silverman, A.I. (2024) FloodNet: Low-cost ultrasonic sensors for real-time measurement of hyperlocal, street-level floods in New York City. *Water Resources Research*, 60, e2023WR036806. https://doi.org/10.1029/2023WR036806

Possible sources of error in flood event and time series data

FloodNet sensors can also measure the distance to non-flood related objects on city streets and sidewalks if placed under the sensor (e.g., people, animals, bicycles, vehicles, garbage, piled snow, and other objects unrelated to flooding). These data may show up as anomalous points or sections of data within the “time series depth data”. Data processing steps are in place to analyze the “time series depth data”. to reduce these non-flood data points in the array, however anomalous data may be observed occasionally. An example of this may be a single anomalous data point, measured during a flood event which may produce an exaggerated value for the “Maximum Flood Depth” metric, and skew the “Time to Maximum Flood Depth” and “Time to Drain From Peak” metrics. Further, anomalous data points that have been filtered from the data set will result in missing time points in the time series data. This can also occur when, on occasion, the sensor sends out a ping, but does not receive an echo back - in these cases, the measurement is invalid and we record a null value.

Why isn't the sensor detecting a flood?

Ultrasonic sensors must be perpendicular to and directly above the location that they are measuring. As such, their ability to measure floods is highly dependent on available sensor mounting locations at locations where flood water is expected to accumulate. FloodNet sensors require existing street infrastructure, such as sign posts, for installation. We do our best to install each sensor near the lowest elevation point at a given flood-prone location, but our reliance on existing infrastructure can limit the locations available for installation, and can result in sensors being installed in locations at a slightly higher elevation than the lowest point on the street, where flood water would accumulate first. Additionally, most FloodNet sensors are mounted over sidewalks and are thus unable to measure floods in the roadbed until the water rises above the height of the curb.

Due to the variable topography of streets and sidewalks, the depth measurements captured by a sensor at a point location will not reflect the depth at every part of the street. For example, if a flood sensor is installed at a location that is 4 inches higher than the lowest point in the area, the sensor would not measure the depth of floods less than 4 inches deep, and a reading of 10 inches would correspond to 14 inches of depth at the lowest point. The "Sensor Height Above Local Low Point (inches)" field in the "Data-Dictionary: Floodnet - Sensor Deployment Metadata.csv" attachment provides the difference between the local low point elevation and the elevation directly beneath the sensor, allowing the flood depth at the local low point to be estimated.

A sensor may also fail to capture a flood event if it was offline and awaiting repair or had not been deployed yet or retired. The "Date Installed" and "Date Removed" fields in the "Data-Dictionary: Floodnet - Sensor Deployment Metadata.csv" attachment can be used to determine a sensor's deployment period. Additionally, a sensor suffering from weak connectivity can fail to transmit measurement data which can result in too few data points to confidently identify a flood for inclusion in this dataset.

During winter weather events, snow can pile up beneath sensors which can lead to erroneous, non-flood measurements. Snow piles can also lead to the sensor missing flood events if the flood depth does not exceed the snow pile depth.

Precipitation and tidal data

If you want to compare flood data with precipitation and tide data, we recommend the following sources:

- **Precipitation** - NYS Mesonet and NYC Micronet weather station data available at 5 minute resolution from 23 stations across the 5 boroughs
- **Tidal** - NOAA & USGS tide gauge data available at 6 minute resolution from stations across the country

Issues in working with the data

If you have any issues with the data, please create an issue on this GitHub repository. Include as much information as possible to help us resolve the issue.

If you have feedback or questions please also post an issue labeled “enhancement” or “question”.

Please reach out to `info@floodnet.nyc` with any other questions. Thanks for your interest and support of our project.